

ELEC473 Assignment 4 - Synthesising the NIOS II Processor

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Abstract

Chapter 1

Introduction

We focus on expanding familiarity of FPGA *FieldProgrammableGateway* engineering through this assignments focus on QSYS [?] and the NIOS-II [1] processor. The NIOS-II being Altera's [?] own 32-bit embedded processor mirco-architecture specifically designed for syntethesis on FPGA, The NIOS-II is a successor of Altera's first configurable 16-bit embedded processor Nios, introduced in 2000. During this assignment we utilize the Cyclone-II DE2 FPGA which we syntethesis designs, bench test and design specifically for during the course of our work.

The NIOS-II hosts the following design parameters: - 32 general-purpose 32-bit registers - Full 32-bit Instruction set, data path, and address space. - Singla-instruction 32 x 32 mulitply and divide producing a 32-bit result.

This soft-core *Asoftmicroprocessorisamicroprocessorcorethatcanbewhollyimplementedusingl*
endandcommodityvalidations [2] nature of NIOS-II enables hardware engineers to specify and generate custom NIOS II core, tailed for their own cusom application requirements.

Qsys (now known as platform designer) is a system integration tool for Intel Quartus Prime. It helps connect multiple intellectual property (IP) blocks, processors (like Nios II), and memory interfaces using standard interconnects like Avalon and AXI. It can be utilized for generated HDL.

This assignment sees us in Part A sythesise a NIOS II processor with 20KB of on-chip memory, a timer, a JTAG UART, 8 parallel I/O pins and system ID component. part a sees us interface the 512 KB SRAM and 8 MB SDRAM on the Altera DE2 Board to this design and test it using memory test programs within NIOS II IDE we will show Name and ID numbers in terminal each time the memory is testing.

part b continues on part a requiring us to develop a custom instruction to count the number of leading 1s in the 32 bit number passed into the instruction. and write a custom program to the this custom instruction. I will also develop a test routine in C or assembler utilizing compiler that has the same instruction

functionality implemented to compare the speed of the custom instruction against software implementation.

Part C we shall utilize PWM to control the intensity of LEDR10.

Chapter 2

Part A - SRAM & SDRAM

Chapter 3

Part B - Custom Instruction

Chapter 4

Part C - PWM Module

Bibliography

- [1] Wikipedia contributors. Nios ii. https://en.wikipedia.org/wiki/Nios_I, 2026. *Accessed* : 2026 – 07 – 31.
- [2] Wikipedia contributors. Soft microprocessor. https://en.wikipedia.org/wiki/Soft_microprocessor, 2026. *Accessed* : 2026 – 07 – 31.